

Given: Due:	Further Maths (Core Pure) HW 01	Name:
Chapter 1 & 2: Complex Numbers & the Argand Diagram		
91% (A*)	76% (A)	66% (B)
56% (C)	50% (D)	40% (E)
A complex number z is given by $z = a + 2i$ where a is a non-zero real number. (a) Find $z^2 + 2z$ in the form $x + iy$ where x and y are real expressions in terms of a. (4) Given that $z^2 + 2z$ is real, (b) find the value of a. (1) Using this value for a, (d) Show the points P, Q and R, representing the complex numbers z, z^2 and $z^2 + 2z$ respectively, on a single Argand diagram with origin O. (3) (e) Describe fully the geometrical relationship between the line segments OP and QR. (2)		

Many students gained full marks for part (a). Mostly correct expressions for $z^2 + 2z$ were seen, bar the odd error. The most common loss of marks was caused by a failure to separate real and imaginary parts, as requested in the question.

Some confusion was evident in part (b) showing, in some cases, a minority erroneously equating the real part to zero. The majority did equate their imaginary part to zero and find $a = -1$.

The approach of most students was correct in part (c) and those with a correct answer to part (b) were usually successful. If a mistake was made in part (b) students would usually achieve one mark for getting their P in the correct quadrant, sometimes gaining the follow through for their R . The majority having produced a correct part (b) went on to get full marks on this part.

Many students spotted that the lines were parallel in part (d), but failed to notice that QR was twice the length of OP . A minority did spot an enlargement with scale factor 2 but very few of them progressed to a correct translation.



Question Number		Marks
	(a) $z^2 = (a+2i)(a+2i) = (a^2-4) + 4ia$ So $z^2 + 2z = (a^2-4+2a) + i(4a+4)$ or $x = (a^2+2a-4)$ and $y = 4a+4$ (b) and so $4a+4 = 0 \rightarrow a = -1$ ALT (b) Substitute $a = -1$ and show that $y = 0$ (c) $z = \sqrt{5}$ or awrt 2.24 $\arctan(-2) = 2.03$ (d) (e) OP and QR are parallel, and QR is twice the length of OP Or Enlargement with Scale Factor 2 (centre O), followed by translation $\begin{pmatrix} -3 \\ -4 \end{pmatrix}$ Or Enlargement with Scale Factor 2, centre (3,4) or centre $3+4i$ $\overrightarrow{QR} = 2\overrightarrow{OP}$ with clear indication of vectors award B1B1, without vectors award B0B1	M1 M1 A1 A1 (4) B1 (1) B1 B1 M1, A1 cao (3) M1 A1 B1ft (3) B1, B1 (2) 13 marks
	Notes: (a) M1: Squares z to produce at least 3 terms which can be implied by the correct answer. M1: Adds $2z$ to their z^2 A1: Correct x A1 Correct y accept $4ai+4i$ (b) B1: Completely accurate cao (c) B1: $\sqrt{5}$ or 2.24 or awrt 2.24 M1 for using tan or arctan A1 cao 2.03 (d) M1: Either their OP in the correct quadrant labelled P or z or their $-1+2i$ or their $(-1,2)$ or axes labelled or their OQ in the correct quadrant labelled Q or z^2 or their $-3-4i$ or their $(-3,-4)$ or axes labelled A1: Both OP and OQ correct i.e. in the 2nd and 3rd quadrants respectively. B1ft: $OR: z^2 + 2z (= -5)$ on real axis to left of the origin. Accept points or lines. Arrows not required. Axes need not be labelled Re and Im. Treat correct quadrant (or on axis) as important aspect for accuracy, lengths of lines if present can be accepted as correct.	

The complex numbers z and w are given by

$$z = 8 + 3i, w = -2i$$

Express in the form $a + bi$, where a and b are real constants,

(a) $z - w$,

(1)

(b) zw .

(2)

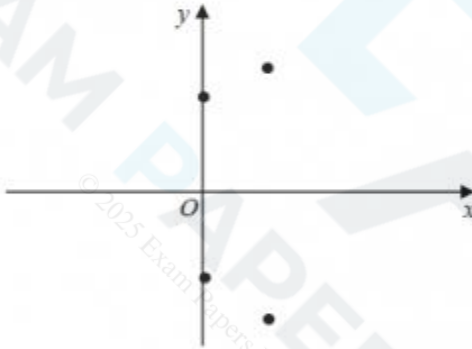
This was a good opening question for the candidates and solutions were almost always correct. They demonstrated that they knew how to deal with real and imaginary parts when subtracting and multiplying very well.

Question Number	Scheme	Marks
(a)	$z = 8 + 3i, w = -2i$ $z - w = (8 + 3i) - (-2i) = 8 + 5i$	B1 [1]
(b)	$zw = (8 + 3i)(-2i) = 6 - 16i$ Either the real or imaginary part is correct.	M1 A1 [2] 3

$f(x) = (4x^2 + 9)(x^2 - 2x + 5)$	
(a) Find the four roots of $f(x) = 0$	(4)
(b) Show the four roots of $f(x) = 0$ on a single Argand diagram.	(2)

Generally both parts of this question were done very well. Errors that candidates made included $\pm \frac{3}{2}$ rather than $\pm \frac{3i}{2}$ or use of an incorrect quadratic formula. Almost all the candidates were able to plot their solutions on a correct Argand diagram.



Question Number	Scheme	Marks
(a)	$f(x) = (4x^2 + 9)(x^2 - 2x + 5) = 0$ $(4x^2 + 9) = 0 \Rightarrow x = \frac{3i}{2}, -\frac{3i}{2}$ $(x^2 - 2x + 5) = 0 \Rightarrow x = \frac{2 \pm \sqrt{4 - 4(1)(5)}}{2(1)}$ $\Rightarrow x = \frac{2 \pm \sqrt{-16}}{2}$ $\Rightarrow x = 1 \pm 2i$	An attempt to solve $(4x^2 + 9) = 0$ which involves i. $\frac{3i}{2}, -\frac{3i}{2}$ M1 Solves the 3TQ A1 1 ± 2i M1 [4]
(b)		Any two of their roots plotted correctly on a single diagram, which have been found in part (a). B1ft Both sets of their roots plotted correctly on a single diagram with symmetry about $y = 0$. B1ft [2] 6
	Method mark for solving 3 term quadratic: 1. <u>Factorisation</u> $(x^2 + bx + c) = (x + p)(x + q)$, where $pq = c$, leading to $x =$ $(ax^2 + bx + c) = (mx + p)(nx + q)$, where $pq = c$ and $mn = a$, leading to $x =$ 2. <u>Formula</u> Attempt to use <u>correct</u> formula (with values for a, b and c). 3. <u>Completing the square</u> Solving $x^2 + bx + c = 0$: $\left(x \pm \frac{b}{2}\right)^2 \pm q \pm c$, $q \neq 0$, leading to $x = \dots$	

$$z = 2 - i\sqrt{3}$$

(a) Show z on an Argand diagram and find the angle the line Oz makes with the positive Real axis.

(3)

Use algebra to express

(b) $z + z^2$ in the form $a + bi\sqrt{3}$, where a and b are integers,

(3)

In part (a) the majority of candidates could at least calculate the magnitude of the angle concerned but there were many cases where the sign was omitted. There were very few cases where candidates worked in degrees. In attempting $z + z^2$ in part (b) most candidates made sound attempts at z^2 although there were some instances of poor algebra and sometimes insufficient work to show that $i^2 = -1$ but on the whole, the correct answer was seen very frequently.

Question Number	Scheme	Notes	Marks
(a)	$\arg z = -\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$	$\tan^{-1}\left(\pm\frac{\sqrt{3}}{2}\right)$ or $\tan^{-1}\left(\pm\frac{2}{\sqrt{3}}\right)$ seen or evaluated	M1
	Awrt ± 0.71 or awrt ± 0.86 can be taken as evidence for the method mark. Or ± 40.89 or ± 49.10 if working in degrees		
	$= -0.7137243789.. = -0.71$ (2 dp)	awrt -0.71 or awrt 5.57	A1
	NB $\tan\left(\frac{\sqrt{3}}{2}\right) = 1.18$ and $\tan\left(\frac{2}{\sqrt{3}}\right) = 2.26$ and both score M0		
			[2]
(b)	$z^2 = (2 - i\sqrt{3})(2 - i\sqrt{3})$ $= 4 - 2i\sqrt{3} - 2i\sqrt{3} + 3i^2$	An attempt to multiply out the brackets to give four terms (or four terms implied).	M1
	$= 2 - i\sqrt{3} + (4 - 4i\sqrt{3} - 3)$ $= 2 - i\sqrt{3} + (1 - 4i\sqrt{3})$ $= 3 - 5i\sqrt{3}$ (Note: $a = 3, b = -5$.)	M1: An understanding that $i^2 = -1$ and an attempt to add z and put in the form $a + bi\sqrt{3}$	M1A1
		A1: $3 - 5i\sqrt{3}$	
	$z + z^2 = 2 - i\sqrt{3} + (4 - 4i\sqrt{3} + 3) = 9 - 5i\sqrt{3}$ scores M1M0A0 (No evidence of $i^2 = -1$)		
			[3]

(c)	$\frac{z+7}{z-1} = \frac{2-i\sqrt{3}+7}{2-i\sqrt{3}-1}$	Substitutes $z = 2 - i\sqrt{3}$ into both numerator and denominator.	M1
	$= \frac{(9-i\sqrt{3})}{(1-i\sqrt{3})} \times \frac{(1+i\sqrt{3})}{(1+i\sqrt{3})}$	Simplifies $\frac{z+7}{z-1}$ and multiplies by $\frac{\text{their } (1+i\sqrt{3})}{\text{their } (1+i\sqrt{3})}$	dM1
	$= \frac{9+9i\sqrt{3}-i\sqrt{3}+3}{1+3}$	Simplifies realising that a real number is needed in the denominator and applies $i^2 = -1$ in their numerator expression and denominator expression.	M1
	$= \frac{12+8i\sqrt{3}}{4}$		
	$= 3+2i\sqrt{3}$ (Note: $c = 3, d = 2.$)	$3+2i\sqrt{3}$	A1
			[4]
(d)	$w = \lambda - 3i$, and $\arg(4 - 5i + 3w) = -\frac{\pi}{2}$		
	$(4 - 5i + 3w = 4 + 3\lambda - 14i)$		
	So real part of $(4 - 5i + 3w) = 0$ or $4 + 3\lambda = 0$	States real part of $(4 - 5i + 3w) = 0$ or $4 + 3\lambda = 0$	M1
	So, $\lambda = -\frac{4}{3}$	$-\frac{4}{3}$	A1
			[2]
Allow $\pm \left(\frac{14}{3\lambda + 4} \right) = \pm \infty \Rightarrow 3\lambda + 4 = 0 \Rightarrow \lambda = -\frac{4}{3}$ A1			
			11 marks

$$f(x) = (x^2+4)(x^2+8x+25)$$

(a) Find the four roots of $f(x) = 0$.

(5)

(b) Find the sum of these four roots.

(2)

A substantial minority multiplied out the two brackets which complicated the problem. Most however attempted to solve $x^2 + 4 = 0$, but there were a number of wrong answers, particularly the real answers +2 and -2. The solution of the three term quadratic was usually correct but there were errors in simplification with a substantial number of candidates losing some accuracy in part (a). Indeed the answers $-4 + 6i$ and $-4 - 6i$ were fairly common. In part (b) candidates were asked to find the sum of their roots. Most obtained -8, but this only gained M1A1 if it followed wrong roots. There were candidates who were unfamiliar with the term "sum" and found the product instead.



Question Number		Marks
Q (a)	$x^2 + 4 = 0 \Rightarrow x = ki, \quad x = \pm 2i$ Solving 3-term quadratic $x = \frac{-8 \pm \sqrt{64 - 100}}{2} = -4 + 3i \text{ and } -4 - 3i$	M1, A1 M1 A1 A1ft (5)
(b)	$2i + (-2i) + (-4 + 3i) + (-4 - 3i) = -8$ Alternative method : Expands $f(x)$ as quartic and chooses $\pm$ coefficient of x^3 -8	M1 A1cso (2) [7] M1 A1 cso
Notes	(a) Just $x = 2i$ is M1 A0 $x = \pm 2$ is M0A0 M1 for solving quadratic follows usual conventions, then A1 for a correct root (simplified as here) and A1ft for conjugate of first answer. Accept correct answers with no working here. Do not give accuracy marks for factors unless followed by roots. (b) M1 for adding four roots of which at least two are complex conjugates and getting a real answer. A1 for -8 following correct roots or the alternative method. If any incorrect working in part (a) this A mark will be A0	